Four-Link Bicopter Aerial Manipulator: Expected Results

Mohammed Abdelsabour · simulated from the single-bicopter flight results · November 2026

Table 1. Simulated configuration

Item	Value used in the simulation
Structure	4 links of 0.56 m, 3 yaw joints; one bicopter at the centre of each link, its arm across the link (rotors beside the link, tilted along it)
Module (from the tested bicopter)	0.80 kg, 2 rotors of 8.2 N max. (hover throttle 0.63 measured), MG996R tilt ±40 deg, 100 Hz, 9 deg hover tilt (CoG offset)
Whole robot	3.74 kg with joint servos and link tubes; hover at 56 % thrust
Sensor and disturbance	1 deg level error per IMU, 3 % rotor mismatch, random air torque
Decentralised	every module runs the single-bicopter firmware on its own IMU
Centralised	one controller, wrench allocation over all 8 rotors (as DRAGON / SPIDAR)

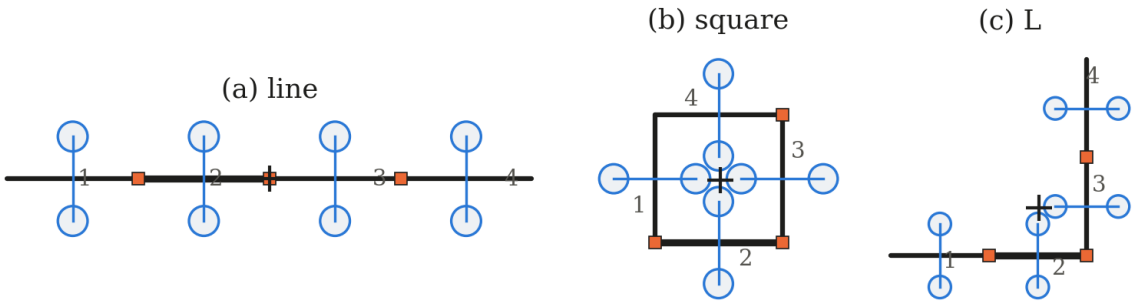


Figure 1. The four-link robot in the line, square and L shapes. Each link carries one bicopter at its centre, rotors across the link; squares are the joints, + the centre of gravity, link 2 the base link.

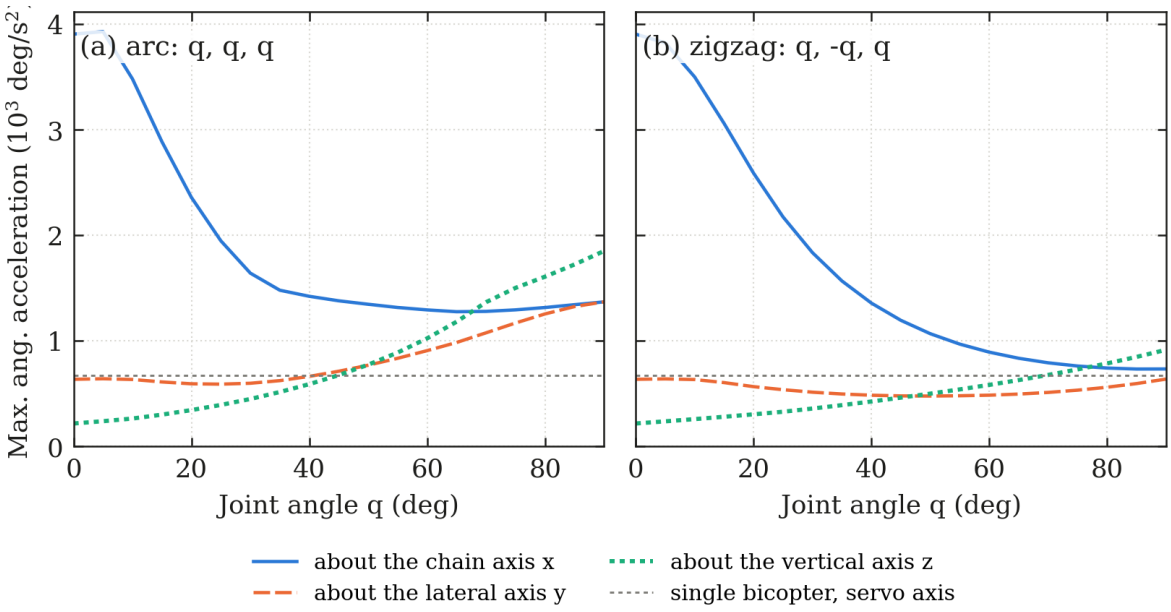


Figure 2. Largest angular acceleration while hovering versus joint angle. The line shape is strongest about the chain axis (3905 deg/s²) and weakest in heading (215 deg/s²); bending trades between them. The single bicopter's servo axis (669 deg/s²) is the grey line.

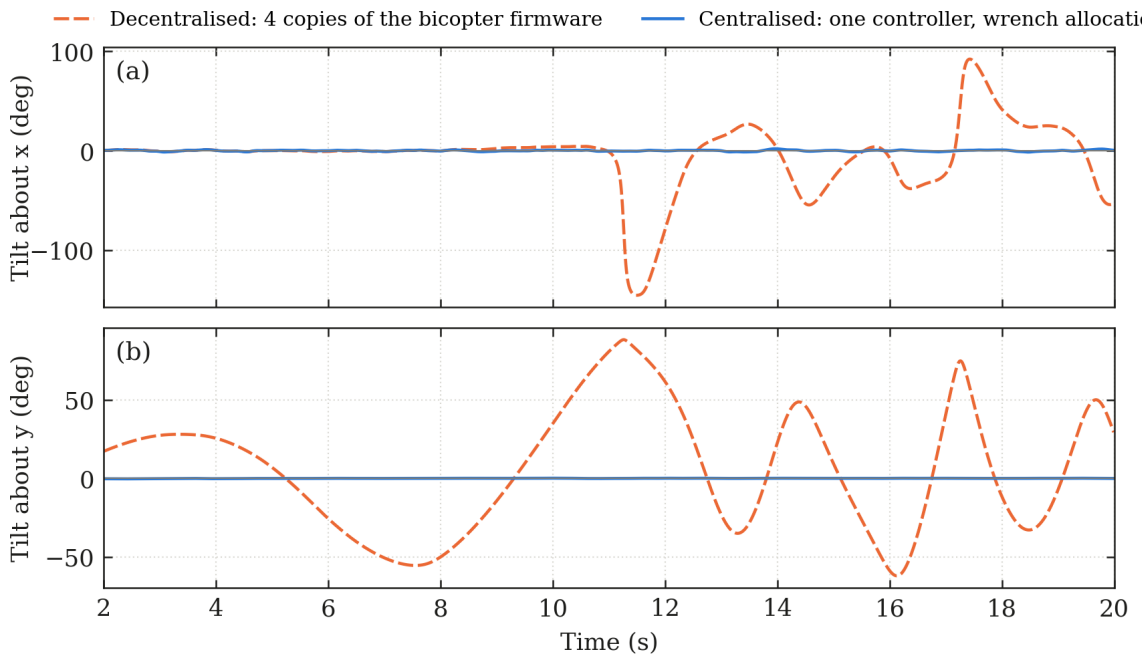


Figure 3. Line shape, 20 s hover. Four independent firmwares lose control (53 deg SD, flips); one central controller holds 0.6 deg SD.

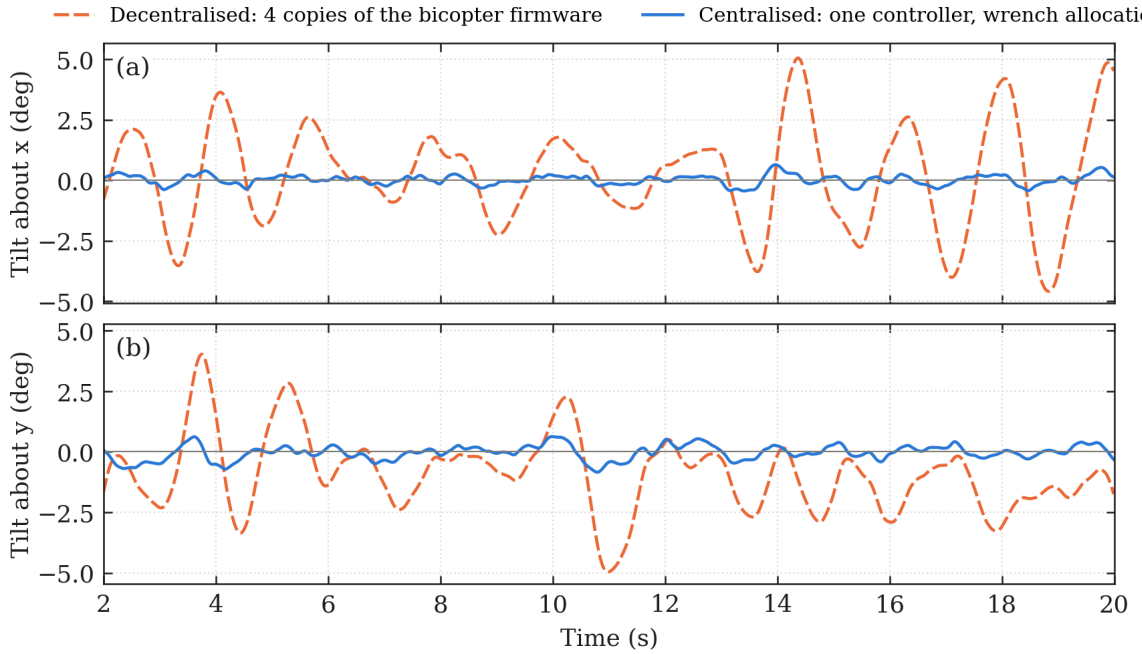


Figure 4. Square shape, 20 s hover. Decentralised 2.6 deg SD, centralised 0.4 deg SD.

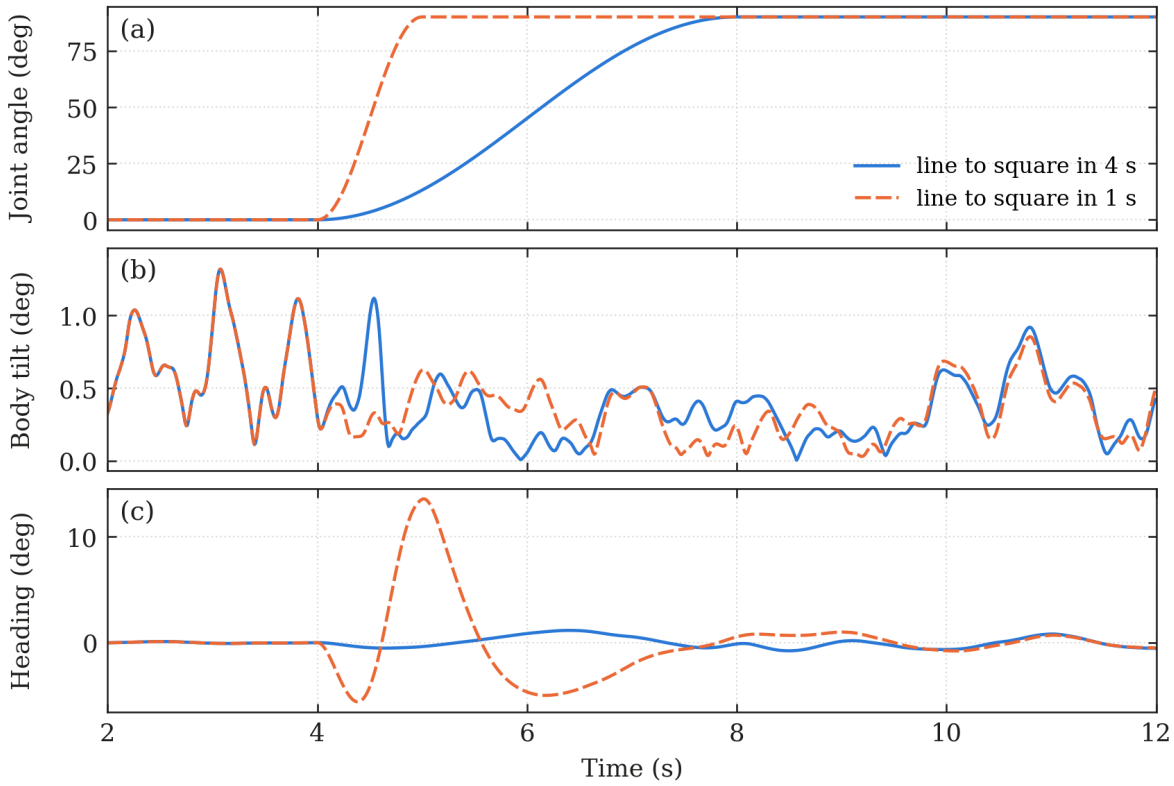


Figure 5. Line to square with the central controller. Tilt stays below 1.1 deg; a 1 s change swings the heading by 19 deg (joint reaction), a 4 s change by 2 deg.

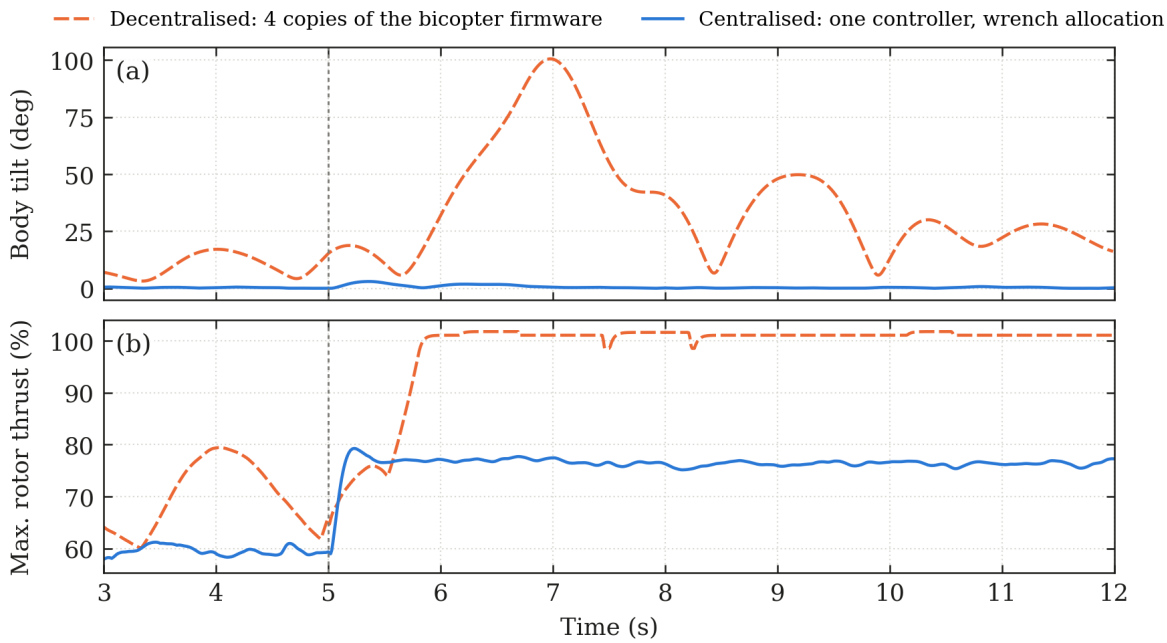


Figure 6. 0.3 kg picked up at the tip of the L shape at t = 5 s. Centralised: 0.9 deg SD, highest rotor 79 %. Decentralised: rotors saturate and the robot tips over (42 deg SD).

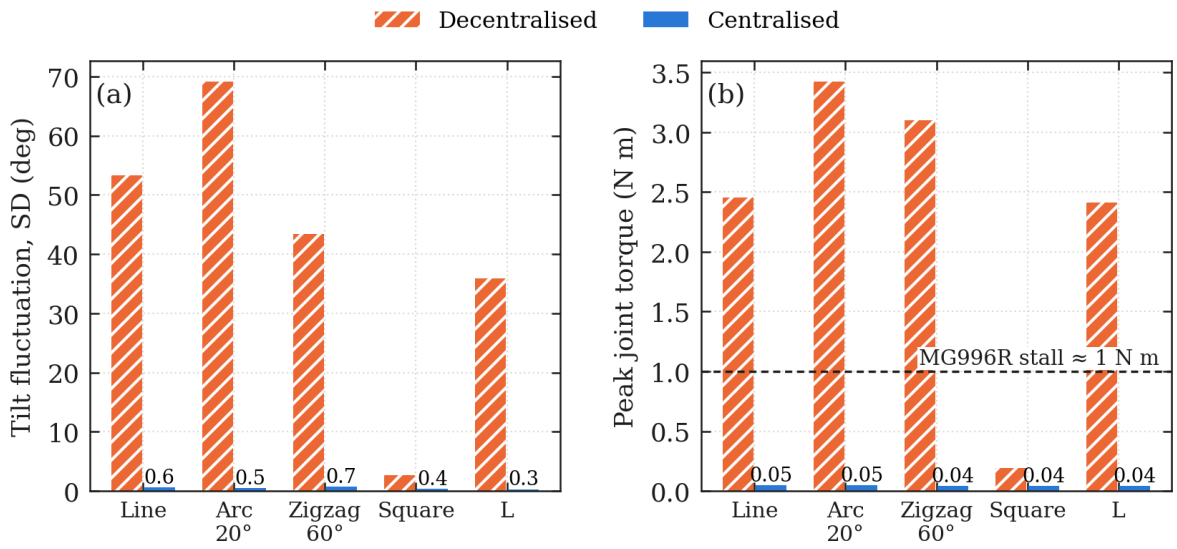


Figure 7. Summary of the hover simulations: (a) tilt fluctuation and (b) peak joint torque in five shapes.

Table 2. Simulated hover results (first 3 s excluded; payload case after the pick-up)

Shape	Controller	Tilt SD (deg)	Mean lean (deg)	Rotor thrust (%)	Joint torque (N m)
Line	decentralised	53.3	7.3	24-100	2.45
	centralised	0.6	0.0	52-60	0.05
Arc 20 deg	decentralised	69.1	20.2	16-100	3.42
	centralised	0.5	0.0	51-61	0.05
Zigzag 60 deg	decentralised	43.4	2.6	12-100	3.10
	centralised	0.7	0.0	51-62	0.04
Square	decentralised	2.6	0.9	36-77	0.19
	centralised	0.4	0.0	49-63	0.04
L	decentralised	35.8	1.1	16-100	2.41
	centralised	0.3	0.0	50-62	0.04
L + 0.3 kg at tip	decentralised	41.6	19.0	16-100	2.53
	centralised	0.9	0.0	41-79	0.03

Table 3. Expected results for the graduation robot

Expected result	**Value**
4 bicopters, each with its own firmware	not flyable except in the square shape (2.6 deg SD); line 53 deg SD, rotors saturate
4 bicopters, one central controller, hover	0.3-0.7 deg SD in every shape, no lean
Strongest / weakest axis in the line shape	chain axis 3905 deg/s² / heading 215 deg/s²
Comparison with the single bicopter (servo axis 669 deg/s²)	line shape: chain axis 5.8 times, lateral axis 0.9 times; square: 2.0 times on both tilt axes
Shape change line to square	4 s: 1.1 deg peak tilt, 2 deg heading swing; 1 s: 0.9 deg, 19 deg
Joint torque	≈ 0.05 N m in hover; up to 2.7 N m in a 1 s shape change (above an MG996R)
Payload	0.3 kg at the tip: 0.9 deg SD, highest rotor 79 %; max. ≈ 0.9 kg with 30 % thrust reserve
Hover thrust	56 % of maximum (thrust-to-weight 1.8)